



Federal Office
for Information Security

Technical Guideline TR-03122-1

Conformance Test Specification for BSI TR-03121 Biometrics for Public Sector Applications

Part 1: Framework

Version 7.0



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1 Changelog

The following tables present the changes introduced to this Technical Guideline since the last released version. The changelog lists the changes grouped per part (test cases) of this Technical Guideline:

- *Added* for new features
- *Changed* for changes in existing functionality
- *Deprecated* for soon-to-be removed features
- *Removed* for now removed features
- *Fixed* for any bug fixes
- *Security* in case of vulnerabilities

1.1 Changelog Version 7.0-draft1

This chapter includes all changes between Version 6.0 and Version 7.0-draft1.

1.1.1 Changelog BSI TR-03122, Part 1

Name	Type of Change	Change Description
-	-	-

Table 1.1 Changelog BSI TR-03122, Part 1

1.1.2 Changelog BSI TR-03122, Part 2

Name	Type of Change	Change Description
TC-HLBS-SRV-ALL-acquireService-003	Changed	Correction of Typos
TC-HLBS-SRV-ALL-acquireService-004	Changed	Correction of Typos
TC-HLBS-SRV-ALL-acquireService-005	Changed	Correction of Typos
TC-HLBS-SRV-ALL-acquireService-006	Changed	Correction of Typos
TC-HLBS-SRV-ALL-beginService-Execution-001	Changed	Correction of Typos
TC-HLBS-SRV-ALL-beginService-Execution-003	Changed	Correction of Typos
TC-HLBS-SRV-ALL-getServiceDescription-001v1	Changed	Correction of Typos
TC-HLBS-SRV-BFIAS-getAllServices-001v1	Changed	Update to TR Version 6.0.
TC-HLBS-SRV-BFIAS-getService-Feedback-001v1	Changed	Addition of details of the response (specified root element).
TC-HLBS-SRV-BFIAS-getService-Feedback-002v1	Changed	Addition of details of the response (specified root element).
TC-TC-HLBS-SRV-FIAS-getAllServices-001v1	Changed	Update to TR Version 6.0.
TC-HLBS-SRV-FIAS-getService-Feedback-001	Changed	Specified the schema to use.

Table 1.2 Changelog BSI TR-03122, Part 2

1.1.3 Changelog BSI TR-03122, Part 3

Name	Type of Change	Change Description
TC-AS-FI-ICS2-007	Changed	Changed format to "bmp". Fixed Colourspace to 24bit. Formalization of instructions.
TC-AS-FI-ICS2-008	Changed	Changed format to "bmp". Fixed Colourspace to 24bit. Refinement and formalization of instructions and Expected Results.
TC-COD-ALL-ARE-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of instructions and Expected Results.
TC-COD-ALL-BCL-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of instructions and Expected Results. Added check for optional elements.
TC-COD-ALL-GID-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Expected Results. Correction of precondition: the connected FP-Scanner is not necessarily a MF scanner.
TC-COD-ALL-IMA-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Expected Results. Added check for optional elements.
TC-AH-ALL-SSS-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of instructions and Expected Results.
TC-AH-FI-BCL-001	Changed	Formalization of Expected Results.
TC-AH-FI-BCL-002	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of Expected Results.
TC-AH-FI-BCL-003	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Expected Results. Definition of "optical path".
TC-AH-FI-BCL-004	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Expected Results. Definition of "optical path".
TC-AH-FI-FBS-001	Changed	Formalization of Preconditions.
TC-AH-FI-FBS-002	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Expected Results. Addition of lower and upper bounds for image resolution. Added Expected Results describing the specific content of the response.
TC-AH-FI-SSS2-004	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of Expected Results (e.g. by adding XPath).
TC-AH-FP-SSS-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of Expected Results (e.g. by adding XPath). Addition of check for image content.
TC-AS-FI-DC-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of instructions and Expected Results (e.g. by adding XPath).
TC-AS-FI-DC2-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of instructions and Expected Results (e.g. by adding XPath).
TC-AS-FI-FBS-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of instructions and Expected Results (e.g. by adding XPath and details on compression).
TC-AS-FI-ICS3-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of instructions and Expected Results (e.g. by adding XPath). Addition of Expected Results regarding the overall result.
TC-AS-FI-ICS3-002	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of instructions and Expected Results (e.g. by adding XPath). Addition of Expected Results regarding the configured timeout and overall result.
TC-AS-FP-MF-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Restructuring of the steps (added configuration steps). Refinement and formalization of Expected Results (e.g. by adding XPath).
TC-AS-FP-ROLL-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement and formalization of instructions and Expected Results (e.g. by adding XPath).

Name	Type of Change	Change Description
TC-AS-FP-SF-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Restructuring of the steps (added configuration steps). Refinement and formalization of Expected Results (e.g. by adding XPath's).
TC-AS-FP-SLP-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Formalization of Expected Results.
TC-AS-FP-SLP-003	Changed	Update w.r.t. to schema trbio5v1.xsd. Formalization of Expected Results.
TC-BIP-FI-FBS-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of Expected Results (e.g. addition of lower and upper bounds for image resolution).
TC-BIP-FI-GID-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of Expected Results (e.g. addition of XPath's, lower and upper bounds for image resolution etc.).
TC-BIP-FP-APP-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Addition of applicability (SF Sensor). Refactoring of Expected Results.
TC-BIP-FP-APP-002	Changed	Update w.r.t. to schema trbio5v1.xsd. Addition of applicability (MF Sensor). Addition of Evaluation Step. Refactoring of Expected Results.
TC-BIP-FP-APP-003	Changed	Update w.r.t. to schema trbio5v1.xsd. Addition of applicability (MF Sensor). Addition of Evaluation Step. Refactoring of Expected Results.
TC-QA-FI-ARE-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. merge of the steps, checks depend on the respective request-image, refinement of the Expected Results, addition of Evaluation Step, Update of DB description). Addition of error rate.
TC-QA-FI-BCL-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. merge of the steps, checks depend on the respective request-image, refinement of the Expected Results, addition of Evaluation Step, Update of DB description). Addition of error rate.
TC-QA-FI-GID-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. merge of the steps, checks depend on the respective request-image, refinement of the Expected Results, addition of Evaluation Step, update of DB description). Addition of error rate.
TC-QA-FI-GID-002	Added	Added TC as an alternate entry point where thresholds for scanned images apply. Addition of error rate.
TC-QA-FI-IMA-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. merge of the steps, checks depend on the respective request-image, refinement of the Expected Results, addition of Evaluation Step, update of DB description). Addition of error rate.
TC-QA-FI-GENERIC-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. merge of the steps, checks depend on the respective request-image, refinement of the Expected Results, addition of Evaluation Step, update of DB description). Addition of error rate.
TC-QA-FI-GENERIC-002	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. merge of the steps, checks depend on the respective request-image, refinement of the Expected Results, addition of Evaluation Step, update of DB and request description). Addition of error rate.
TC-QA-FP-APP-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. update of instructions, refinement of the Expected Results, addition of Evaluation Step, update of DB and request description). Addition of error rate.
TC-QA-FP-APP-002	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. update of instructions, refinement of the Expected Results, addition of Evaluation Step, update of DB and request description). Addition of error rate.
TC-QA-FP-APP-003	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. update of instructions, refinement of the Expected Results, addition of Evaluation Step, update of DB and request description). Addition of error rate.
TC-QA-FP-APP-004	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. update of instructions, refinement of the Expected Results, addition of Evaluation Step, update of DB and request description). Addition of error rate.

Name	Type of Change	Change Description
TC-QA-FP-APP-005	Added	TC was added as an alternate entry of TC-QA-FP-APP-003 for IUT that only work on 500ppi images.
TC-COD-FI-GENERIC-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Expected Results.
TC-COD-FI-VER-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Expected Results.
TC-COD-FP-VER-001	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Expected Results.
TC-PAD-FI-APP-002	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. merge of steps, update of instructions, addition of evaluation step).
TC-PAD-FP-APP-002	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of the entire TC (e.g. merge of steps, update of instructions, addition of evaluation step).
TC-PAP-ACQ-FI-AUTO-1-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Steps, instructions and Expected Results.
TC-PAP-ACQ-FI-AUTO-1-002v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Steps and Expected Results.
TC-PAP-ACQ-FI-SV-2-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Re-ordering and refinement of Steps and Expected Results. Added more details regarding provision of timestamps.
TC-PAP-ACQ-FI-SV-2-002v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Re-ordering and refinement of Steps and Expected Results. Added more details regarding provision of timestamps. Exclusion of BMP.
TC-PAP-ACQ-FI-SV-2-003v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Re-ordering and refinement of Steps and Expected Results.
TC-PAP-ACQ-FI-SV-4-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Steps and Expected Results. Addition of test resources.
TC-PAP-ACQ-FI-SV-5-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Steps and Expected Results. Added applicability and deleted the prefix to the step ID.
TC-PAP-ACQ-FI-SV-5-002v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Steps and Expected Results. Added test resources.
TC-PAP-ACQ-FI-SV-5-003v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Steps and Expected Results. Added test resources. Changed the last part of the URI.
TC-PAP-ACQ-FP2P-SV-2-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of Steps and Expected Results. Addition of test resources in Preconditions. Split the last Expected Results into smaller parts. Added XPath expressions. Differentiated cases where either the left or right index was marked as "missing" or captured with insufficient quality.
TC-PAP-ACQ-FP2P-SV-2-002v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of Steps and Expected Results. Addition of test resources in Preconditions. Split the last Expected Results into smaller parts. Added XPath expressions.
TC-PAP-ACQ-FP2P-USV-1-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring of Steps and Expected Results. Addition of test resources and Preconditions. Replaced underscore in TC-identifier.
TC-PAP-ACQ-FP2P-USV-2-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refinement of Steps (added configuration in Step 2, specified which fingers to use etc.) Expected Results. Addition of test resources and Preconditions. Replaced underscore in TC-identifier.
TC-PAP-ACQ-FP10R-SV-1-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring and refinement of Steps, instructions and Expected Results.
TC-PAP-ACQ-FPS-USV-1-001v1	Changed	Update w.r.t. to schema trbio5v1.xsd. Refactoring and refinement of Steps, instructions and Expected Results. Addition of test resources and configuration steps.
TC-AH-FI-DC-001	Changed	Formalization of Expected Results and Applicability.

Name	Type of Change	Change Description
TC-AH-FI-DC-002	Changed	Exclude the usage of grey scale for facial verification. Formalization of Applicability.
TC-AH-FI-DC2-001	Changed	Formalization of Applicability.
TC-AH-FI-DC2-002	Changed	Formalization of Applicability.
TC-AH-FI-ICS-001	Changed	Formalization of Preconditions and Applicability.
TC-AH-FI-ICS-002	Changed	Formalization of Preconditions.
TC-AH-FI-ICS-003	Changed	Exclude the usage of grey scale for facial verification. Formalization of Preconditions.
TC-AH-FI-ICS-004	Changed	Formalization of Preconditions.
TC-AH-FI-ICS2-001	Changed	Formalization of Preconditions and Applicability.
TC-AH-FI-ICS2-002	Changed	Formalization of Preconditions.
TC-AH-FI-ICS2-003	Changed	Exclude the usage of grey scale for facial verification. Formalization of Preconditions.
TC-AH-FI-ICS2-004	Changed	Formalization of Preconditions and Applicability.
TC-AH-FI-ICS2-005	Changed	Formalization of Preconditions and Applicability.
TC-AH-FI-SSS-001	Changed	Formalization of Preconditions.
TC-AH-FI-SSS2-001	Changed	Formalization of Preconditions.
TC-AH-FI-SSS2-002	Changed	Formalization of Preconditions.
TC-AH-FI-SSS2-003	Changed	Formalization of Preconditions.
TC-AH-FP-ALL-001	Changed	Refactoring of Expected Results.
TC-AH-FP-OPT-001	Changed	Refactoring of Expected Results and Applicability.
TC-AH-FP-OPT-002	Changed	Refactoring of Expected Results.
TC-AH-FP-OPT-003	Changed	Refactoring of Expected Results.
TC-AH-FP-OPT-004	Changed	Refactoring of Expected Results.
TC-AH-FP-OPT-005	Changed	Formalization of Preconditions.
TC-AS-FI-ICS-004	Changed	Exclude the usage of grey scale for facial verification.
TC-AS-FI-ICS-004	Changed	Exclude the usage of grey scale for facial verification.
TC-AS-FI-ICS2-004	Changed	Exclude the usage of grey scale for facial verification.
TC-BIP-FI-APP-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources. Exclude the usage of grey scale for facial verification.
TC-AH-FI-DC-003	Added	New TC
TC-AH-FI-DC-004	Added	New TC
TC-AH-FI-EGT-001	Changed	Formalization of Preconditions.
TC-AH-FI-EGT-002	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-BIP-FI-DC-HQ-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-COM-FI-BCL-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-COM-FI-IMA-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-COM-FI-JPG-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.

Name	Type of Change	Change Description
TC-COM-FP-BCL-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-COM-FP-IMA-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-COM-FP-WSQ-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-COM-FP-WSQE-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-CMP-FP-CC-001	Changed	Renamed to TC-CMP-FP-CV-001
TC-CMP-FI-GENERIC-001	Deleted	Deleted TC as the FM is not used.
TC-PAP-ACQ-FP4141-SV-1-001	Changed	Refactoring and refinement of scope, instructions, Expected Results and test resources.
TC-PAP-ACQ-FP442-SV-1-001	Changed	Refactoring and refinement of scope, instructions, Expected Results and test resources.
TC-PAP-ACQ-FP2P-SV-1-001	Changed	Refactoring and refinement of Steps, instructions, Expected Results and test resources.
TC-PAP-ACQ-FP2P-SV-1-002	Changed	Refactoring and refinement of instructions, Expected Results and test resources.
TC-PAP-ACQ-FP10R-SV-1-002	Changed	Refactoring and refinement of instructions, Expected Results and test resources.
TC-PAP-ACQ-FPS-SV-1-001	Changed	Refactoring and refinement of instructions, Expected Results and test resources.
TC-PAP-ACQ-FPS-SV-1-002	Changed	Refactoring and refinement of instructions, Expected Results and test resources.
TC-PAP-ACQ-FPS-SV-1-003	Changed	Refactoring and refinement of instructions, Expected Results and test resources.
TC-PAP-ACQ-FPS-SV-1-004	Changed	Refactoring and refinement of instructions, Expected Results and test resources.
TC-PAP-ACQ-FPS-SV-1-005	Changed	Refactoring and refinement of instructions, Expected Results and test resources.

Table 1.3 Changelog BSI TR-03122, Part 3

1.2 Changelog Version 7.0-draft2

This chapter includes all changes between Version 7.0-draft1 and Version 7.0-draft2.

1.2.1 Changelog BSI TR-03122, Part 1

Name	Type of Change	Change Description
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Table 1.4 Changelog BSI TR-03122, Part 1

1.2.2 Changelog BSI TR-03122, Part 2

Name	Type of Change	Change Description
-	-	-

Table 1.5 Changelog BSI TR-03122, Part 2

1.2.3 Changelog BSI TR-03122, Part 3

Name	Type of Change	Change Description
TC-PAP-DEL-FI-SV-1	Added	Added TC by using fragments of the TC-PAP-ACQ-FI-SV-2.
Tests_PAP	Changed	Change due to addition of the TC-PAP-DEL-FI-SV-1.
TC-COD-FI-ROD-001	Changed	Fixed the image resolution to 1244x1600 pixels.
Second Addendum TR03122	Added	New Addendum for background elimination.
TC-LOG-ALL-GID-001	Changed	Moved parts to new TC-LOG-ALL-BDS-001.
TC-LOG-ALL-BDS-001	Added	Created new TC with parts of TC-LOG-ALL-GID-001.
TC-O-ALL-USV-001	Added	Created new TC as placeholder.

Table 1.6 Changelog BSI TR-03122, Part 3

1.3 Changelog Version 7.0

This chapter includes all changes between Version 7.0-draft2 and Version 7.0.

1.3.1 Changelog BSI TR-03122, Part 1

Name	Type of Change	Change Description
Schema	Changed	Introduce new schema version 6v1 for all volumes.

Table 1.7 Changelog BSI TR-03122, Part 1

1.3.2 Changelog BSI TR-03122, Part 2

Name	Type of Change	Change Description
-	-	-

Table 1.8 Changelog BSI TR-03122, Part 2

1.3.3 Changelog BSI TR-03122, Part 3

Name	Type of Change	Change Description
TC-AS-FP-ROLL-003	Changed	Changed CTS mode.
TC-LOG-ALL-ARE	Changed	Specified which root element and sub-elements to expect.
TC-LOG-ALL-BCL	Changed	Specified which root element and sub-elements to expect.
TC-LOG-ALL-GID	Changed	Specified which root element and sub-elements to expect.
TC-PAP-ACQ-FI-SV-2	Deleted	Deleted the test case from the list.
TC-AS-FI-ICS2-009	Changed	Added one section with respect to remaining parts of the background.
TC-AH-FI-FBS-002	Changed	Deleted tolerance of acquisition resolution.
TC-BIP-FI-FBS-001	Changed	Deleted tolerance of image resolution.
TC-LOG-FI-GENERIC	Changed	Added a remark with respect to cancelled acquisitions.
TC-LOG-FP-GENERIC	Changed	Added a remark with respect to cancelled acquisitions.
TC-LOG-FI-GENERIC	Added	Added the test case TC-LOG-FI-GENERIC-002.
TC-UI-FI-OP	Changed	Changed wording and added requirements in the case of a PAD alarm.
TC-O-ALL-USV	Added	Removed placeholder and added three test cases.

Name	Type of Change	Change Description
TC-AS-FI-ICS3-002	Changed	Replaced "non-compliant face presentations" by the term "leaving the capture area".
TC-QA-FI-GID-001, TC-QA-FI-GID-002	Changed	Fixed reference to FM QA-FI-GID.
TC-LOG-ALL-ARE	Changed	Deleted the logging of a cancelled process.
TC-AS-FI-DC2-001	Changed	Added one bullet point for Description and one for Expected Result - both regarding background of the image (for Step 1).
TC-BIP-FI-GID-001	Changed	Added (as an expected result for step 1) that padding has not been used during image processing.
TC-BIP-FI-DC-HQ-001	Changed	Added (as an expected result for step 1) that padding has not been used during image processing.
TC-PAP-ACQ-FI-AUTO-1-001	Changed	Excluded background replacement from the required time for the facial image acquisition in GID.

Table 1.9 Changelog BSI TR-03122, Part 3

2 Introduction

This document is part one of the Conformance Test Specification (BSI TR-03122). It is the counterpart of BSI TR-03121-1 describing the framework of the conformance test documents for BSI TR-03121.

2.1 Motivation and Objectives of the Conformance Test Specification

The Technical Guideline Biometrics for Public Sector Applications (BSI TR-03121) specifies requirements and recommendations for the use of biometric data within the scope public sector applications. The requirements on specific Function Modules, as defined in BSI TR-03121-3, can be implemented for different public sector applications through hardware and software components from various vendors.

The objective of this Technical Guideline is to offer a base for consistent and comparable quality assessment regarding the different components that will be applied in order to fulfil these requirements. This conformance test specification

- specifies tests for the Software Architecture (in particular in regard to conformance testing),
- defines all test cases being relevant to verify the conformance for the different requirements described in the Function Modules.

2.2 Target Audience

Audience for this guideline are institutions that are dealing with projects using biometrics in public sector applications that require certified modules, hardware, and/or software. These include:

- Vendors of hardware or software products that want to present their solutions for conformance test and acquire to be compliant to this Technical Guideline.
- Evaluation laboratories that check the conformance of hardware and/or software modules that are used within the scope of biometrics and electronic identity documents in public sector applications.

2.3 Structure of the Conformance Test Specification

The Conformance Test Specification consists of the following parts:

- Part 1: Framework (BSI TR-03122-1)

The BSI TR-03122-1 is the framework document of the conformance test specification.

- Part 2: Test Cases for High Level Biometric Services (HLBS) (BSI TR-03122-2)

The second part defines test cases for servers and clients implementing the High Level Biometric Services interface.

- Part 3: Test Cases for Function Modules and Processes (BSI TR-03122-3)

The third part defines test cases for hardware and software components as well as processes according to their specification in TR-03121-3.

2.4 Test Case Specification for Function Modules

1. The requirements for an electronic identity document and the connected public sector application are combined in several Function Modules within BSI TR-03121-3. If the conformance to one or more specific Function Modules has to be checked, the reader has to identify at first the relevant Function Modules with the according abbreviation e.g. LOG-FP-GID.
2. As a result the corresponding test cases can be selected in BSI TR-03122-3 under the same identification while the prefix 'TC-' indicates the according test case. Depending on the number of defined test cases the abbreviation is followed by an ascending numbering e.g. TC#LOG#FP-GID-001.
3. In general, a test case in BSI TR-03122-3 is structured in three parts as shown in ►Table 2.1.

Test Case ID: TC-LOG-FP-GID-001	
Scope	
Short overview of the test case	
Preconditions	
Requirements that need to be fulfilled before the test case can be executed	
Description	
Enumerated listing of every single test step	
Expected Result	Description of the expected result for the corresponding test step

Table 2.1 Example of the Structure of a Test Case

Conformance to the specification of a Function Module can be established if all test cases for that Function Module are completed successfully and the requirements of the test methodology are satisfied.

Several test cases can be assigned to one Function Module each containing one or multiple test steps.

Note, that the Conformance Test Specification does not define requirements for the object to be tested except the interfaces for conformance testing.

2.5 Conformance Test Interfaces

Conformance testing in the context of BSI TR-03121 is based on an interface compliant to Representational State Transfer (REST). This interface is described in the following sections.

The Implementation Under Test (IUT) must provide an external interface as REST service accessible to the Conformance Test Suite (CTS), through which conformance testing is performed. Using this interface, the IUT is able to output generated data to the CTS during the test execution of the regular process which is performed by the operator of the evaluation laboratory.

Additionally, the interface can be used to provide pre-defined input from the CTS to the IUT in order to be able to verify its output in a defined state.

In opposite to the standard workflow, this request is used as an alternative point of entry so that the standard steps for the acquisition of the biometric image (e.g. facial image or fingerprint image), further processing of the image and/or compression of the image can be skipped.

In case pre-defined input data is required for a test case, the IUT is provided with external test data through the interface by the CTS. The input data is provided as XML data with root element “biocets-testsetup” as defined in the XML schemata of BSI TR-03121. The schema definition can be found in the file “biocets6v1.xsd”. An example can be found in the file “biocets-testsetup.xml”.

2.5.1 Interface description

In the conformance test scenario, a client-server architecture is at hand. The IUT represents the server providing a HTTP-based REST-interface to the CTS which acts as the client.

A test case is triggered by an HTTP request sent by the CTS to the IUT. The path as part of the request Uniform Resource Identifier (URI) specifies the test case ID as defined in part 3 of this guideline:

```
/TR03122/{testcase-id}/{version}
```

The version indicates the revision of the specification for this test case and is typically incremented when the test case’s interface requirements are changed.

The HTTP request method (relevant are GET and POST) depends on whether the test case requires input data. As described above, any input data for the IUT must be embedded in XML data (biocets-testsetup) within the request.

The result generated by the IUT must also be returned as XML data to the CTS. Depending on the test case at hand, the result consists of XML data conforming to BSI TR-03121, which further may include an embedded, application specific format (e.g. German Standard for AFIS Transactions Version 3 (GSAT3) XML). The specific

data format and URI for test case initiation is described further in each test case definition of BSI TR-03122 Part 3.

Since performing a test case may comprise manual interaction within the IUT and hence can take accordingly long, the communication timeout has to be considered and set adequately high.

2.5.1.1 Test Cases Not Requiring Pre-Defined Input Data

In case a test case does not require any input data a priori, the CTS sends a GET request to the IUT. The IUT is triggered upon receiving the request and performs the test case (e.g. facial image capture and encoding).

Subsequently, the result data is returned to the CTS within the response body. The response header must be set to "Content-Type: application/xml; charset=utf-8" and include the message length of the response body.

Example:

Request (CTS → IUT):

```
GET /TR03122/TC-COD-PH-GSAT3-001/1 HTTP/1.1
```

```
Content-Type: application/xml; charset=utf-8
```

Response (IUT → CTS):

```
HTTP/1.1 200 OK
```

```
Content-Type: application/xml; charset=utf-8
```

```
Content-Length: 12345
```

```
<are:aad-app [...]
```

```
<bio:Records>
```

```
<bio:XMLRecord type="gsat-xml" purpose="enrolment" id="id_1234" size="5687">
```

```
[...]
```

```
</are:aad-app>
```

2.5.1.2 Test Cases Requiring Pre-Defined Input Data

Should a test case require initial provision of data, the CTS sends a POST request to the IUT including this data. The request header must be set to "Content-Type: application/xml; charset=utf-8" and include the content length. Any input data is provided within the message body as XML (UTF-8 encoded).

The result data is returned to the CTS within the response body including a response header set to "Content-Type: application/xml; charset=utf-8" and respective content length.

Example:

Request (CTS → IUT):

```
POST /TR03122/TC-QA-PH-SB-001/1 HTTP/1.1
```

```
Content-Type: application/xml; charset=utf-8
```

```
Content-Length: 12345
```

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
[...]
```

```
<biocts:Parameter type="face" format="bmp">
```

```
VghpcyBmaWVQLg==</biocts:Parameter>
```

```
</biocts:biocts-testsetup>
```

Response (IUT → CTS):

```
HTTP/1.1 200 OK
```

```
Content-Type: application/xml; charset=utf-8
```

Content-Length: 67890

<?xml version="1.0" encoding="UTF-8"?>

<FaceQuality [...]

2.5.1.3 Response Codes

►Table 2.2 lists the possible HTTP status codes which must be returned by the IUT to indicate the status to the CTS.

Status Code	Status Message	Description
200	OK	The request was successfully processed. Test case result data is included in the response body.
400	Bad Request	The IUT could not process the request due to defective input data.
404	Not Found	The IUT cannot perform the test case, e.g. due to missing implementation.
500	Internal Server Error	An internal, technical error occurred in the IUT during processing the request.

Table 2.2 Possible Status Codes Used by the IUT

List of Abbreviations

Abbreviation	Description
CTS	Conformance Test Suite
GSAT3	German Standard for AFIS Transactions Version 3
IUT	Implementation Under Test
REST	Representational State Transfer
URI	Uniform Recourse Identifier

